



BY ELECTRONIC SUBMISSION VIA [REDACTED]

March 15, 2025

Faisal D'Souza
Networking and Information Technology Research and Development (NITRD)
National Coordination Office (NCO)
2415 Eisenhower Avenue
Alexandria, VA 22314, USA

RE: AI Action Plan RFI (FR Doc: 2025-02305)

Dear Mr. D'Souza,

As an established leader in the use of artificial intelligence (AI) and machine learning (ML) to accelerate the development of transformative products across the life sciences landscape, Flagship Pioneering respectfully submits these comments in response to the National Science Foundation's Request for Information on the Development of an AI Action Plan.

By way of background, Flagship Pioneering creates, resources, and develops original bioplatfrom companies that aim to transform human health and sustainability, utilizing advanced biological technologies and processes, AI/ML, and novel manufacturing technologies. Over the last 25 years, Flagship Pioneering has created a broad ecosystem of pioneering bioplatfroms and companies, collectively representing:

- More than 100 scientific ventures founded, including more than a dozen publicly-traded companies;
- 28 FDA authorized, approved, or cleared products;
- More than 190 product candidates in development (from vaccines to drugs against cancer, and for rare and common diseases);
- an annual, collective R&D budget of more than \$2 billion;
- more than 10,500 employees across our companies from Cambridge to Seattle to San Francisco to Indiana, with the vast majority working directly on R&D; and
- 3,100,000+ square feet of laboratory and manufacturing space across the United States.

Flagship boasts a decades-long history of leveraging cutting-edge AI/ML-enabled bioplatfroms to create transformative biotech companies and products. We are at the forefront of harnessing exponential advancements in Generative AI, reinforcement and multimodal machine learning, large language models, and AI-enhanced simulations. These technologies empower us to pioneer new knowledge and decode the languages of life, enabling our scientists and entrepreneurs to accelerate and refine the process of scientific hypothesis ideation, iteration, and testing. This approach allows us to develop groundbreaking products that redefine the biotech landscape. For example, our proprietary AI/ML systems enable us to:

- Model disease processes to learn how cells move from health to disease, and then computationally identify diverse sets of molecules that are capable of reversing or stopping these processes;
- Computationally generate, design, and optimize novel protein therapeutics (biologics/biotherapeutics) and new chemical entities (small molecule drugs) with desired functions

and properties, and do so faster, cheaper and better than traditional scientific methods, which are reliant on inefficient trial-and-error;

- Identify optimal small molecule drugs based on history of safe human use;
- Identify optimal small molecule drugs that are synthesized by human commensal microbes and have potent bioactivities;
- Experimentally and computationally optimize DNA- or RNA-based medicines for expression, stability, and lack of immunogenicity, in order to expand the utility of these medicines beyond vaccines;
- Computationally elucidate biological patterns that correspond to cancer pathways for the early and accurate detection of oncogenesis; and
- Leverage AI/ML to innovate across the scientific method using scientific superintelligence and autonomous labs.

In addition, Flagship is building and constantly refining a proprietary, centralized AI-driven intelligence engine to systematize our knowledge and augment our capabilities across our portfolio of bioplatforms and companies. We also are establishing partnerships across the AI landscape with leading technology developers to further power breakthrough innovations in AI and the life sciences.

Based on this experience and leadership in the field, Flagship Pioneering respectfully submits the following comments and recommendations:

SUMMARY OF COMMENTS

Leadership in AI/ML – and particularly within the context of biotechnology innovation – is critical to the United States’ future national and economic security. These technologies enhance defense capabilities by improving intelligence, threat detection, and cybersecurity strategies, and are vital for ensuring public health preparedness against biological threats – whether intentional or naturally occurring. They also enhance our economic growth and competitiveness by enabling the United States to remain at the cutting-edge of technological and efficiency advances. Increased federal funding for AI/ML research and the expansion of public-private partnerships will further accelerate innovation and deployment. Through focused investment and well-crafted policy measures that effectively link technological supremacy with economic prosperity and national security, the U.S. can safeguard its national interests, preserve its strategic advantage amid rapid global technological and geopolitical changes, and advance the interests of humanity worldwide.

To build a robust national infrastructure that unleashes waves of American-led AI/ML innovation, we recommend the U.S. government focus on the following priorities:

- **Coordinated Management, Sharing, and Generation of Biological Data to Propel Innovation:** Establish coordinated management and sharing of biological data across the federal government’s vast array of agencies, to help standardize protocols and improve access to high-quality, AI-ready datasets for public and private researchers. Invest in innovative data generation technologies to maintain U.S. leadership in AI.
- **Fostering AI Innovation and Leadership through Limited, Risk-Based Regulation:** Foster private investment in AI through limited, risk-based regulatory frameworks. Shape consistent global standards, particularly in drug development, to enhance and streamline responsible innovation for patients globally.

- **Restrict U.S. Outbound AI Investments with Sanctioned Countries:** Regulate outbound investments in AI, including research and development activities, with sanctioned entities to protect national security, safeguard critical technology sectors, and sustain technological leadership.
- **Incentivizing AI Innovation and Expanding Access to Expertise:** Uphold strong patent protections for AI-enabled inventions to encourage innovation and U.S. leadership. Create mechanisms to access foreign talent. Establish a specialized AI/ML team within the FDA to streamline regulatory processes for AI/ML use in drug development.

SPECIFIC COMMENTS

The federal government must act decisively to establish a robust infrastructure that will propel the development of AI and ML technologies forward and unleash waves of American-led innovation. These technologies are not only pivotal in shaping the future economic and national security of the United States, but also are critical in addressing some of the most pressing challenges of our time. The United States needs a comprehensive framework that unites the diverse capabilities and resources of the public and private sectors behind this collective mission. This framework must, at its core, focus on enhancing and integrating diverse datasets vital for training state-of-the-art AI and ML models, which are foundational to driving breakthroughs in vital sectors such as healthcare, national defense, energy, and agriculture.

This framework also should envision strategic public-private investments to support the building of needed AI infrastructure, along the lines of the critical infrastructure sovereign wealth fund idea proposed by President Trump and other experts in the field. The lab facilities and equipment needed to advance such innovations are often well beyond the financial capacity of the private sector, particularly among the many small businesses that are leading research in this area. The federal government can play an important role in catalyzing strategic financing in this sector.

The overarching goal should be to create an AI-powered future that enhances economic prosperity, improves the quality of life for Americans, and ensures access to technological benefits across all sectors of U.S. society, guaranteeing that AI's integration is not only beneficial but truly transformative. The following sections describe the key pillars of this necessary infrastructure in more detail. By channeling dedicated resources and attention into these areas, the U.S. government can promote sustainable economic growth and reinforce the U.S.'s competitive edge in the global technology race.

I. Coordinating Management, Sharing, and Generation of Biological Data to Propel Innovation

The federal government should establish mechanisms for coordinated management and sharing of biological data across its vast network of agencies, and frameworks for sharing such data with responsible public and private researchers with appropriate privacy protections. This initiative is crucial for standardizing data protocols that ensure consistency and interoperability among a wide array of datasets, unlocking the full potential of interdisciplinary research. By implementing uniform approaches, this effort will enable the private and public sector research communities to access and utilize data with unprecedented efficiency, nurturing a collaborative research ecosystem that transcends existing institutional and disciplinary boundaries.

By making high-quality, AI-ready, comprehensive datasets more accessible, the U.S. can spark transformative developments across the life sciences, especially in areas such as disease prevention, cell & gene therapy, energy efficiency, and agricultural innovation. Furthermore, by reducing the cost and time associated with

disaggregated data acquisition and “clean up,” the initiative will significantly boost economic growth in strategic sectors and maximize the potential for revolutionary discoveries that will shape the future – cementing the U.S.’s position as a global leader in biotechnology and AI.

Furthermore, the U.S. should prioritize and significantly increase investment in advancing data generation capabilities, which are crucial for capturing comprehensive biological insights. These include innovative technologies focused on the deep measurement of biology and human physiology at scale. The ability to create scalable and renewable sources of biological data will decisively shape the competitive landscape in the coming years as advancements in AI capabilities will naturally gravitate toward regions that offer abundant and rich data resources. These pioneering technologies are vital for maintaining the United States’ leadership in AI, biotechnology, and scientific research.

II. Fostering AI Innovation and Leadership through Risk-Based Regulation

The federal government should focus on fostering an environment that encourages continued private investment to unleash the full potential of AI by avoiding regulations that single out AI models for burdensome and likely unworkable restrictions. While we acknowledge that AI, like any technology, can be used for good or for harm, the government should focus on identifying or preventing actual harmful activities rather than, for example, regulating AI based solely on a model’s computing power or other features. Such regulations, as mentioned in previous executive orders and legislative discussions, could inadvertently stifle innovation and deter private companies from investing in advanced AI capabilities. In the face of increasing competition from abroad, where centralized and well-funded AI initiatives are rapidly advancing, it is crucial that the U.S. maintains a competitive edge. This requires an approach that incentivizes investment and supports the scalable growth of AI technologies domestically, rather than imposing blanket limitations that could hinder progress.

Instead of focusing on computing capacity or model size, the federal government should emphasize a risk-based regulatory framework that considers the intended use of AI models. This approach would allow for more nuanced governance that addresses potential risks without stifling innovation. Encouraging public-private cooperation will be paramount in developing these frameworks, as it offers a platform for sharing insights, expertise, and best practices. By promoting a collaborative environment, the government can help ensure AI technologies are developed responsibly and ethically, while also supporting commercial growth and maintaining global competitiveness.

Moreover, the U.S. should actively engage on the international stage to shape global AI regulatory frameworks that encourage and support innovation. In drug development, for example, AI can accelerate research processes, enhance target identification, and optimize clinical trials, leading to faster and more effective therapies for patients worldwide. As drug development continues to expand globally, such international cooperation on AI can help establish consistent, risk-based regulatory environments conducive to both progress and patient protection.

III. Protecting U.S. Leadership in AI Innovation through Strong Patent Policies

The federal government should prioritize the protection and encouragement of innovation within the AI sector by ensuring that AI-enabled inventions remain patentable. Recently adopted guidance by the U.S. Patent and Trademark Office (USPTO) under the last Administration could create uncertainty over this question, as it appears to impose new limits or requirements on AI-related patents on the theory that machines, rather than humans, are the inventors or co-inventors of the invention. Such restrictions could

deter investment in AI research and development, as companies and inventors may lack confidence in their ability to secure intellectual property rights. Such an approach also could disadvantage the United States in the global technology race, as competitor nations have, to date, not sought to treat AI-enabled inventions any differently than other tools used by humans in the inventive process – thus potentially driving the most advanced AI innovation out of the United States.

Maintaining a robust patent system that accommodates AI-driven innovations is essential for fostering continued U.S. leadership in AI and technology. The USPTO should rescind its misguided guidance and reaffirm the central role that humans play in designing, managing, directing, monitoring, and evaluating the output of AI models to bring valuable new products to the market. This will ensure that U.S.-based researchers and companies remain at the forefront of technological advancements, and that the U.S. will remain in a position to promote similar patent policies abroad for American inventors.

Furthermore, in order to incentivize the private sector to leverage federal investments in data generation and data sharing, the federal government must ensure that private sector companies feel secure in the ownership of their intellectual property generated from using government-provided datasets. Clarifying the scope of federal rights, such as march-in rights, for such IP will alleviate concerns that these rights might be exercised too broadly. By establishing clear guidelines and assurances around IP ownership, the government can foster a more collaborative environment that encourages private investment and innovation while leveraging public data assets effectively.

IV. Restricting U.S. Outbound AI Investment and Research with Sanctioned Countries

The federal government should continue to fortify robust restrictions on U.S. outbound AI investments and research collaborations with companies in sanctioned countries to safeguard national security and maintain technological leadership. We support the U.S. Department of Treasury's recent [Final Rule](#)¹, which, for the first time beyond traditional sanctions, regulates outbound investments for national security reasons. This rule mandates notification or prohibition of certain transactions involving U.S. persons with covered foreign persons engaged in critical technology sectors. By implementing these new regulations faithfully, the government can decisively protect national interests and ensure that U.S. resources are not inadvertently aiding technological competitors.

V. Ensuring Access to AI/ML Expertise

Given the extraordinary, untapped power of AI/ML technologies to rapidly accelerate the design, development, and regulatory review of products that can cure and prevent diseases, the federal government should establish a specialized cross-center team within the Food and Drug Administration (FDA) with designated expertise in AI/ML. This team should serve as an initial advisory entity and central clearinghouse for voluntary pre-Investigational New Drug (pre-IND) inquiries from drug sponsors seeking to use AI/ML in novel ways. The objective is to assist those seeking to integrate AI and ML into drug development by offering a streamlined entry point that enables sponsors to efficiently determine whether their proposed use falls under FDA regulatory oversight. If oversight is applicable, the team would then identify the appropriate office

¹ U.S. Department of the Treasury. (2024). Provisions Pertaining to U.S. Investments in Certain National Security Technologies and Products in Countries of Concern. *Federal Register*, Volume 89, 90398-90473. <https://www.federalregister.gov/documents/2024/11/15/2024-25422/provisions-pertaining-to-us-investments-in-certain-national-security-technologies-and-products-in>



or program within the agency to provide specific guidance on the proposed use, while ensuring consistency across the agency and providing specific expertise as needed.

By consolidating AI and ML inquiries within a single, expert-led team, the FDA can facilitate quicker and more efficient communications with developers, ensuring that innovative technologies are leveraged as fully as possible and without causing unnecessary delays. This team would work in close collaboration with existing review divisions to maintain consistent and expert oversight on queries related to AI/ML use, ensuring that technological integration occurs rapidly and responsibly in alignment with the agency's regulatory standards. This prompt action will not only advance the agency's capacity to manage emerging technologies but also position the U.S. as a leader in the integration of AI and ML in drug development, ultimately benefiting human health and innovation in the biopharmaceutical sector globally.

More broadly, the U.S. government should establish mechanisms to expedite and expand access by American entities to foreign AI/ML talent through reforms to applicable immigration processes, particularly among allied country partners. We need to encourage AI/ML experts to come and remain in the United States to help propel American innovation forward.

CONCLUSION

Flagship Pioneering appreciates the opportunity to provide these comments in response to your Request for Information on the Development of an AI Action Plan. We look forward to further opportunities to engage with you and other stakeholders on this topic. Thank you for your consideration of our views.

Sincerely,

/s/

Tom DiLenge
Senior Partner, Global Public Policy, Regulatory
& Governmental Strategy
Flagship Pioneering